

Nisha Bajiya

Computational Biologist - Artificial intelligence - Peptide and Aptamer Therapeutics

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Profile

Computational biologist with a research focus on peptide and aptamer therapeutics. Experienced in building machine-learning predictors, manually curated databases, and public web servers for antibacterial peptides, cell-penetrating peptides, and anticancer, antiviral, and antibacterial aptamers. Proficient in feature engineering, deep learning, and protein and nucleotide language models (ESM-2, ProtBERT, PepBERT, DNABERT, HyenaDNA, Nucleotide Transformer), with software shared through GitHub and PyPI.

Education

Ph.D. in Computational Biology	9.00/10 CGPA
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India	2022 – 2027
M.Sc. in Biotechnology	91.08% (2019)
Jaipur National University, Jaipur, India	University Gold Medal
B.Sc. in Biotechnology	82.7% (2017)
University of Rajasthan (Biyani Girls College), Jaipur, India	First Class

Publications

Peer-reviewed journal

- **AptViralDB: A repository of experimentally validated antiviral aptamers**
Bajiya N, Singh S, Gahlot PS, Raghava GPS Molecular Therapy - Nucleic Acids, 2026
- **Prediction of inhibitory peptides against *E. coli* with desired MIC value**
Bajiya N, Kumar N, Raghava GPS Scientific Reports, 2025
- **CPPsite3: An updated large repository of experimentally validated cell-penetrating peptides**
Bajiya N, Najrin S, Kumar P, Choudhury S, Tomer R, Raghava GPS Drug Discovery Today, 2025
- **AntiBP3: A method for predicting antibacterial peptides against gram-positive/negative/variable bacteria**
Bajiya N, Choudhury S, Dhall A, Raghava GPS Antibiotics, 2024
- **Advances in the field of phage-based therapy with special emphasis on computational resources**
Bajiya N, Dhall A, Aggarwal S, Raghava GPS Briefings in Bioinformatics, 2023
- **In-silico tool for predicting and scanning rheumatoid arthritis-inducing peptides in an antigen**
Tomer R, Jain S, Gahlot PS, Bajiya N, Raghava GPS Frontiers in Immunology, 2025
- **Prediction of plant resistance proteins using alignment-based and alignment-free approaches**
Gahlot PS, Choudhury S, Bajiya N, Kumar N, Raghava GPS Proteomics, 2024
- **Prediction of anti-freezing proteins from their evolutionary profile**
Kumar N, Choudhury S, Bajiya N, Patiyal S, Raghava GPS Proteomics, 2024
- **MRSLpred: A hybrid approach for predicting multi-label subcellular localization of mRNA at the genome scale**
Choudhury S, Bajiya N, Patiyal S, Raghava GPS Frontiers in Bioinformatics, 2024
- **Multi-perspectives and challenges in identifying B-cell epitopes**
Kumar N, Bajiya N, Patiyal S, Raghava GPS Protein Science, 2023

Preprints

- **AntiCapt: Fine-tuned nucleotide language models for predicting and designing anticancer aptamers**
Bajiya N, Mehta NK, Raghava GPS bioRxiv, 2026
- **AptCancerDB: A curated knowledgebase and translational discovery platform for anticancer aptamers**
Bajiya N, Singh S, Raghava GPS bioRxiv, 2026
- **AptBacterialDB: A comprehensive, manually curated database of antibacterial aptamers**
Bajiya N, Gupta I, Raghava GPS bioRxiv, 2026
- **CPPLocPred: Subcellular localization of cell-penetrating peptides**
Bajiya N, Mehta NK, Raghava GPS bioRxiv, 2026
- **Prediction of lncRNA-protein interacting pairs using LLM embeddings based on evolutionary information**
Choudhury S, Bajiya N, Raghava GPS bioRxiv, 2025
- **In silico tool for identification of colorectal cancer from cell-free DNA biomarkers**
Mathur K, Jain S, Bajiya N, Kumar N, Raghava GPS arXiv, 2025

Book chapters

- **Chapter 19: Machine learning-based methods for designing protein-based drugs**
Bajiya N, Rathore AS, Kumar N, Raghava GPS Royal Society of Chemistry, 2026, vol. 87, pp. 641–681
- **Browsing and mapping of cell-penetrating peptides in the CPPsite3 database**
Bajiya N et al. Under publication
- **Prediction, design, and genome-wide scanning of plant disease resistance proteins using PlantDRPpred**
Raghava GPS et al. Under publication

Research Experience

January 2022 - Doctoral Researcher, IIIT-Delhi, New Delhi

January 2027 *Supervisor: Prof. Gajendra P. S. Raghava*

- Built ML/DL predictors for peptide activity and therapeutic aptamers
- Curated large sequence datasets from literature and public databases
- Developed Python pipelines on HPC systems for large-scale model training
- Applied peptide, protein, and nucleotide language models for feature extraction
- Released web servers, standalone tools, and GitHub/PyPI packages

January 2019 - Dissertation Research (M.Sc.), Jaipur National University, Jaipur (Rajasthan)

June 2019 *Supervisor: Dr. Divya Shrivastava, Dr. Smita Pareek*

- Dissertation on isolation and antifungal susceptibility testing of dermatophytes
- Processed clinical samples from 60 patients; KOH microscopy, SDA/DTM culture
- Identified species by LPCB staining and slide culture; disk diffusion testing

June 2016 - Dissertation Trainee (B.Sc.), Dr. B. Lal Institute of Biotechnology, Jaipur (Rajasthan)

August 2016 *Supervisor: Dr. Parul Chowdhury*

- Dissertation on drought-stress effects on antioxidant enzymes in *Vigna radiata*
- Pot experiment on two mung bean varieties under three drought levels
- Assayed CAT, SOD, APX, and GPX by UV-Vis spectrophotometry; Bradford assay
- Completed plant biotechnology training (1–15 June 2016)

June 2015 - Research Trainee, Biyani Girls College, Jaipur (Rajasthan)

July 2015 *Supervisor: Ms. Heena Kewalramani*

- Completed training on electrophoretic techniques (15 days)

Current Projects

1. Therapeutic Aptamer Prediction

2. Pan-Cancer Genomic Map of Therapeutic Aptamer Targets

Skills

Languages	Python, R, Bash/Linux, SQL (MySQL), HTML/CSS, JavaScript, PHP
Scientific Python and Cheminformatics	NumPy, Pandas, SciPy, Matplotlib, Jupyter Notebook, RDKit
Machine Learning	scikit-learn, XGBoost, random forest, TensorFlow/Keras, PyTorch, CNN, LSTM, cross-validation, feature selection, dimension reduction
Language Models	PepBERT, ProtBERT, BioBERT, ESM-2, DNABERT, HyenaDNA, Nucleotide Transformer
Sequence and Structure Analysis	BLAST(p/n)/PSI-BLAST, CD-HIT, MERCI motif analysis, Pfeature/Nfeature descriptors, Biopython, ViennaRNA
Data Resources	UniProt/Swiss-Prot, GenBank, GDC (Genomic Data Commons)
Cancer Omics	DNA methylation and gene expression in cancer cells
Tool Development	Git/GitHub, PyPI packaging, standalone software, web servers built with MySQL and HTML/CSS/JavaScript, knowledge graphs, manual literature curation
Laboratory	BSL-3 handling, bacterial and fungal culture maintenance, KOH and LPCB microscopy, antifungal disk diffusion, Bradford assay, antioxidant enzyme assays (CAT, SOD, APX, GPX), gel electrophoresis, plant pot experiments
Equipment	Bright field microscope, laminar air flow, UV-Vis spectrophotometer, refrigerated centrifuge, autoclave, hot air oven, pH meter, analytical balance, BOD incubator

Teaching and Mentorship

Teaching Assistant, IIIT-Delhi, New Delhi

- Courses: Machine Learning for Biomedical Applications (two TAs); Big Data Mining in Healthcare (one TAs)
- Ran tutorial sessions on machine-learning methods, ways to assess models, and working with biological data
- Helped students build ML workflows and make sense of biological datasets
- Evaluated assignments and gave written feedback on students' computational work

Student Mentorship

- Guided 3 students through their thesis research on computational biology and ML problems
- Supported data preprocessing, model building, validation, and database development
- Advised on project planning, debugging, interpretation of results, and workflow design

Honors, Fellowships and Achievements

- Dean's List, Best Teaching Assistant (BIO542), IIIT-Delhi, 2023–24
- CSIR-UGC NET Junior Research Fellowship, Life Sciences (2020): **All India Rank 97**
- ICMR Junior Research Fellowship, Life Sciences (2020)
- DBT Junior Research Fellowship (2020)
- GATE 2021, Biotechnology: All India Rank 334
- Qualified the Ph.D. entrance examination, All India Institute of Medical Sciences (AIIMS), New Delhi (2021)
- University Gold Medal for First Position, M.Sc. Biotechnology, Jaipur National University (2020)
- First Position, B.Sc. II Year (Biotechnology), Biyani Girls College, Jaipur (2015–16)

Workshops, Symposia and Training

- Workshop: *Establishing and Sustaining Cloud Infrastructure for Informatics-based Healthcare*, IIIT-Delhi, 22–23 March 2024
- Symposium on Computational Gastronomy, IIIT-Delhi, 10 December 2023
- National Workshop-cum-Training Programme on *Open Source Tools in Bioinformatics*, Birla Institute of Scientific Research, Jaipur, 21–23 January 2015

Extracurricular Activities

- Completed 5 km run at the Faridabad Half Marathon (3 March 2024) and 10 km at the Decathlon 10K Run, Delhi (22 September 2024)
- Second position in Yoga, Karwaan (IIIT-Delhi inter-branch league), 2025; performed at International Yoga Day (June 2026) and Happy Hours (March 2026), IIIT-Delhi
- Voluntary blood donation, IIIT-Delhi camp organized with Hindu Rao Hospital, March 2024

References

Prof. Gajendra P. S. Raghava

Professor

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